Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sanshya 12 of 2011)
University under section 2(f) of UGC Act, 1956

GEHU/04M/9.1.3

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.Tech, ME III semester
Name of the Paper: Material Science
Paper Code: TME-302
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.
(iii) Please specify COs against each question.

Q1. (10 Marks)

a. Classify different engineering materials and their properties. (CO 1)

OR

b. Define the different types of interatomic bond. (CO 1)

Q2. (10 Marks)

a. What are the different defects in crystal? Define. (CO 1)

b. What role does microstructure play in the performance of ceramic materials? (CO 2)

Q3. (10 Marks)

a. Compare the structures of NaCl and CsCl. What are the key differences? (CO 2)

OR

b. What is the significance of fullerenes in nanotechnology? (CO 2)

Q4.

a. How is X-ray diffraction used to determine crystal structures? (CO 2)

OR

b. What are Fick's laws of diffusion and how are they applied in materials science? (CO 2)

Q5. (10 Marks)

a. Write short notes on nanocrystalline solids and polymeric materials. (CO 2)

OR

b. How do grain boundaries influence the mechanical properties of materials? (CO 1)

Note For the question paper setters:

- Question paper should cover all the COs of the course.

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